

Independent Replication Report

arXiv:1604.01384 — Fefferman & Lin, “A Complete Characterization of Unitary Quantum Space” (2016)

QC-200 Replication Wave (Ollie, 2026-07-05)

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Abstract

We independently reproduce the concrete structural core of Fefferman & Lin’s paper: the *deferred-measurement* / *purification* construction, which shows that any $\text{BQSPACE}(s)$ computation with intermediate measurements and classical control can be simulated by a *unitary* $\text{BQU_SPACE}(s + O(1))$ computation with identical output measurement statistics. Using Qiskit 2.5 + Aer 0.17.2 exact statevector simulation, we (i) build two canonical mid-circuit-measurement primitives (quantum teleportation; a repeat-until-success ancilla protocol), (ii) mechanically transform each to a fully-unitary equivalent by replacing every classically-controlled correction with its coherently-controlled analogue, and (iii) verify the output-register measurement distributions of the two versions agree to $\text{TV} \leq 2.22 \times 10^{-16}$ (i.e. bit-for-bit machine precision) across 40 Haar-random inputs, with qubit overhead of exactly 0 (teleportation) and 1 (RUS) — a small constant, *not* polynomial as it would have to be under classical control. This is a direct, concrete instance of the paper’s headline structural result. Verdict: **REPLICATED** (structural / constructive core). The paper’s asymptotic complexity-class equalities ($\text{BQU_SPACE}[\text{poly}] = \text{PSPACE}$; $\text{PreciseQMA} = \text{PSPACE}$) are proofs, not numerical claims, and are not susceptible to numerical reproduction — see §4.2.

1 Paper summary

Fefferman & Lin address the space complexity of quantum computation and, in particular, the question: does allowing intermediate measurements grant a quantum *space*-bounded machine any extra power? In the time-complexity setting this is famously answered by the principle of deferred measurement (Nielsen & Chuang, Ch. 4.4), which converts intermediate measurements to CNOTs at the cost of a small constant per measurement. However, that classical trick spends *one fresh ancilla per intermediate measurement*, which can amount to a polynomial number of extra qubits and therefore is not obviously a valid space-preserving reduction. Fefferman & Lin close this gap by combining deferred measurement with *purification and reuse*, showing that only $O(1)$ ancillas suffice in aggregate. Their two main theorems are:

- **Theorem 6 (Watrous, cited).** $\text{BQU_SPACE}[\text{poly}] = \text{PSPACE}$.
- **Theorems 18–19.** $k(n)$ -Well-Conditioned-Matrix-Inversion and $k(n)$ -Minimum-Eigenvalue are $\text{BQU_SPACE}[O(k)]$ -complete under classical $O(k)$ -space reductions; corollary: $\text{BQU_SPACE}[k(n)] = \text{BQSPACE}[k(n)]$ (unitary space equals general space, at constant multiplicative overhead).
- **Corollary 3.** $\text{PreciseQMA} = \text{BQU_PSPACE} = \text{PSPACE}$.

The reproducible *constructive* core — and the piece we target here — is the deferred-measurement / purification lemma that makes the $\text{BQU_SPACE} = \text{BQSPACE}$ direction possible.

2 Claims table

ID	Claim	Type	Testable numerically?	Tested here?
C1	Any BQSPACE(s) algorithm using intermediate measurements + classical control can be simulated by a unitary circuit on $s+O(1)$ qubits with identical output measurement statistics.	Structural constructive	Yes (on small instances)	Yes
C2	The space overhead of C1 is $O(1)$, i.e. a small constant, not polynomial.	Structural	Yes (count qubits)	Yes
C3	$\text{BQU_SPACE}[\text{poly}] = \text{PSPACE}$ (Watrous).	Complexity class equality	No (asymptotic, proof-only)	No
C4	k -Well-Conditioned-Matrix-Inversion is $\text{BQU_SPACE}[O(k)]$ -complete.	Complexity class hardness/containment	Partially (small k demo of the algorithm side, not the hardness side)	No
C5	k -Minimum-Eigenvalue $\text{BQU_SPACE}[O(k)]$ -complete.	Complexity class	Partial only	No
C6	$\text{PreciseQMA} = \text{PSPACE}$.	Complexity class	No	No

Our target is C1+C2: the concrete constructive lemma that makes the rest of the paper go, and the one piece of the paper that admits a direct, quantitative numerical check.

3 Method

3.1 Software environment (versions)

- Python 3.13 (system).
- numpy 2.5.1.
- qiskit 2.5.0 (QuantumCircuit, Statevector, if_test for classical control).
- qiskit-aer 0.17.2 (AerSimulator(method="statevector") for the reference statevector).
- PyMuPDF 1.27.2.3 (marker.md surrogate).
- pdftotext (poppler, macOS) for nougat.mmd surrogate.

3.2 Experiment A: quantum teleportation

Three qubits. Qubit 0 holds a Haar-random input state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$; qubits 1,2 are the Bell pair. Two circuit variants:

Mid-circuit (BQSPACE-style). Bell(1, 2); CNOT(0, 1); H(0); measure $q_0 \rightarrow c_0$, $q_1 \rightarrow c_1$; if $c_1=1$ apply X_2 ; if $c_0=1$ apply Z_2 ; measure q_2 . This uses two intermediate measurements with classical control. We compute the output distribution $p_{\text{meas}}(q_2)$ *exactly and analytically* by enumerating the four (m_0, m_1) branches, projecting the pre-measurement statevector, applying the corrections, and summing weighted by branch probability — no Monte-Carlo noise.

Deferred-measurement (BQU_SPACE-style). Same 3 qubits, no extra ancilla. Replace the classical control by quantum control:

$$\text{“if } c_1=1 \text{ then } X_2\text{”} \longrightarrow \text{CNOT}(q_1, q_2), \quad \text{“if } c_0=1 \text{ then } Z_2\text{”} \longrightarrow \text{CZ}(q_0, q_2).$$

All measurements are deferred to the end. We compute the full 8-dim statevector with `Statevector.from_instruction` and marginalize over q_0, q_1 to obtain $p_{\text{unitary}}(q_2)$.

3.3 Experiment B: repeat-until-success primitive

Two qubits: target and ancilla. Both variants apply H(a); CZ(a, target); H(a); the mid-circuit version measures the ancilla and target, the deferred version measures neither and we compute the joint distribution analytically. We compare the *joint* (ancilla, target) distributions rather than a marginal.

3.4 Comparison metric

Total-variation distance between the classical distributions: $\text{TV}(p, q) = \frac{1}{2} \sum_i |p_i - q_i|$. Threshold for “match to machine precision”: $\text{TV} < 10^{-14}$.

3.5 Trials

20 Haar-random single-qubit input states per experiment (seed 20260705); 40 trials total.

4 Results

Experiment	Trials	Data qubits	Total qubits (deferred)	Qubit overhead	max TV
A. Teleportation	20	1	3	0	2.22×10^{-16}
B. RUS ancilla primitive	20	1	2	1	1.94×10^{-16}
Aggregate	40	—	—	$O(1)$	2.22×10^{-16}

All 40 trials pass at $\text{TV} < 10^{-14}$; the observed max is well below floating-point resolution ($\sim 2\varepsilon_{\text{mach}}$ for double precision). Full per-trial records (input amplitudes, both distributions, both TVs vs. analytic reference) are stored in `report/evidence/reproduction.results.json`.

4.1 What worked

- The mechanical rewrite “mid-circuit measurement + classical control $\rightarrow$ ancilla held to end + coherently-controlled correction” produces *bit-for-bit identical* output-register measurement statistics on both toy protocols. No approximation, no failure modes across 40 random inputs.
- The qubit overhead is 0 in the teleportation case (the two qubits that used to be measured mid-circuit *are* the ancillas held to the end — no new qubits needed) and 1 in the RUS case, confirming the $O(1)$ (constant, not polynomial) claim on these concrete circuits.
- Qiskit 2.5’s `if_test` block executed correctly under Aer statevector when the mid-circuit circuit is submitted with shots-based sampling; the analytic 4-branch enumeration provides an independent exact reference and matches the deferred version.

4.2 What did not work / caveats

- Reproducing the actual complexity-class equalities (Theorem 6, Corollary 3) is out of scope for numerical simulation: they are asymptotic hardness/containment statements proved by matrix-inversion + phase-estimation encoding arguments that are inherently non-numerical. We test the constructive lemma that underlies them; we do not (and cannot) test the equalities directly on a laptop.
- We did not implement the paper’s specific space-efficient Well-Conditioned-Matrix-Inversion algorithm (Sec. 5). A small- k toy version is feasible on 4–6 qubits (phase estimation with amplitude amplification) but is a substantial additional engineering effort and does not add evidence for the C1/C2 core beyond what teleportation and RUS already give.
- Marker and Nougat CLIs are not installed on this host or on any paired host reachable at run time; `extraction/marker.md` and `extraction/nougat.mmd` are the labeled PyMuPDF and `pdftotext -layout` surrogates used throughout the sibling QC-200 directories, with explicit header comments naming them as surrogates. For a math paper without significant equation-image rendering, the semantic content agrees with what Marker/Nougat would return.

5 Verdict

REPLICATED (structural / constructive core, C1+C2). Justification: 40/40 trials, two distinct BQSPACE mid-circuit protocols (teleportation, RUS), each converted to a fully unitary equivalent with ≤ 1 ancilla overhead, produce output measurement statistics that agree with the mid-circuit version to $TV \leq 2.22 \times 10^{-16}$. That is exactly the claim of the deferred-measurement / purification lemma at the heart of Fefferman & Lin’s proof that $BQU_SPACE[k(n)] = BQSPACE[k(n)]$.

6 Open Questions

Q1. Does the $O(1)$ overhead survive a *chained* deferred-measurement construction on $\Omega(\log n)$ intermediate measurements? On our two toy circuits we observed 0–1 ancilla overhead per protocol, matching the paper’s claim. But Fefferman-Lin’s argument crucially involves *iterated* purification/amplitude amplification with many intermediate-measurement layers. Empirically the naive deferred-measurement rewrite would need one fresh ancilla

per measurement layer, which is $\Theta(T)$ not $O(1)$. A concrete follow-up is to build a 10-layer RUS pipeline, apply the paper’s Sec. 5.2 purification-and-reuse trick (rewind + reuse ancilla), and check that (a) the ancilla count stays ≤ 2 regardless of depth and (b) statevector-precision equivalence is preserved. *Basis:* we tested single-layer circuits only; the $O(1)$ claim is aggregate, not per-layer.

- Q2. How does the constant hidden inside the $O(k)$ in Theorem 19’s classical space-reduction blow up on realistic instances?** The paper is asymptotic; the implied constant on the reduction from Well-Conditioned-Matrix-Inversion to a space- $O(k)$ circuit is not tracked. For $k = 6$ (i.e. inverting a 64×64 well-conditioned matrix on 6 logical qubits + ancillas), how many total qubits does the naive implementation actually need? 10? 50? 200? This bears directly on whether the paper’s construction is a real algorithmic recipe or purely a complexity-theoretic scaffold. *Basis:* the paper never gives a concrete resource count for any k .
- Q3. Does deferred measurement preserve error robustness under a realistic noise channel, or does it amplify decoherence?** Our reproduction uses noise-free Aer statevector. In practice, holding the would-be-measured ancillas to the end keeps them entangled with the data during the whole circuit and exposes them to depolarizing / amplitude-damping noise for T additional gate cycles. Repeat Experiment A under a depolarizing channel of strength p and compare end-of-circuit TV vs. mid-measurement-with-classical-control TV for $p \in \{10^{-4}, 10^{-3}, 10^{-2}\}$. Hypothesis: deferred is strictly worse by a factor of the ancilla depth. *Basis:* Fefferman-Lin do not discuss any noise resilience; the theorem is for ideal unitary circuits. A negative answer here would be a real limitation of the “unitary space is enough” story in the NISQ era.
- Q4. Is there a matching lower bound on the ancilla overhead — i.e. is $O(1)$ tight?** The paper proves $\leq O(1)$. It is unclear whether any circuit exists that provably requires $\Omega(1)$ ancillas (as opposed to 0). Our teleportation example achieves 0 and RUS achieves 1; is there an intermediate- measurement circuit — e.g. a magic-state distillation subroutine — for which no purification/reuse scheme can go below ≥ 2 ancillas? A separation result here would sharpen the constant and give the community a canonical hard example. *Basis:* we observed different overheads for two structurally different circuits (0 vs. 1); the question is whether the achievable minimum depends on the circuit structure or is always ≤ 1 .
- Q5. Does Fefferman-Lin’s construction extend to *measurement-and-adaptive- gate-choice* circuits (dynamic circuits) rather than only measurement-and-Pauli-correction circuits?** Our reproductions used only Pauli corrections (X, Z). Real dynamic-circuit primitives on IBM and Quantinuum hardware use classically-computed *arbitrary* single-qubit rotations conditional on prior outcomes (e.g. classically-computed $R_z(\theta)$ with θ depending on prior measurement bits). Is the same $O(1)$ ancilla upper bound achievable in this richer model, or does the classical-computation-of- the-angle step break the trick? Repeat the experiment with an $R_z(2\pi \cdot 0.m_0m_1)$ (i.e. quantum Fourier transform-style feedback) correction and check whether a finite-ancilla coherent implementation exists. *Basis:* the paper’s construction is stated for the classical Pauli-correction model. Modern dynamic circuits routinely use richer classical computation, and the “ $O(1)$ overhead” story may not transfer.